Solutions: Problem Set 2

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Problem 2.1 (Sample size). We use the data set `sepe.dat` from Problem Set 1.

a) We plan a new sample survey to estimate total environmental protection expenditures. The survey will use simple random sampling (SRS), and the total will be estimated by $\widehat{t}_y = N\bar{y}$, where $\bar{y}$ is the sample mean.

For sample size planning, assume that future expenditures have approximately the same coefficient of variation as the expenditures observed in `sepe.dat`. Determine the sample size so that the relative error exceeds $\rho = 10\%$ with probability at most $\alpha = 5\%$.

For the population mean, the requirement is

$$\mathbb{P}\left(\left|\frac{\bar{y} - \bar{y}_U}{\bar{y}_U}\right| > \rho\right) \leq \alpha.$$

Rewrite this condition for $\widehat{t}_y$. What is the minimum required sample size?

Solution. Since $\widehat{t}_y = N\bar{y}$ and $t_y = N\bar{y}_U$,

$$\left|\frac{\widehat{t}_y - t_y}{t_y}\right| = \left|\frac{\bar{y} - \bar{y}_U}{\bar{y}_U}\right|.$$

Thus, the same relative-error condition applies to the estimator of the total. Using a normal approximation and initially ignoring the finite-population correction, we obtain

$$n_0 = \left(\frac{z_{1-\alpha/2} \cdot \sigma_y}{\rho\, \bar{y}_U}\right)^2 = \left(\frac{z_{1-\alpha/2} \cdot \mathrm{CV}_y}{\rho}\right)^2,$$

where σ_y is the finite-population standard deviation and $\mathrm{CV}_{=}\sigma_y/\bar{y}_U$ is the finite-population coefficient of variation. Accounting for the finite-population correction gives

$$n = \frac{n_0}{1 + n_0/N}.$$

Calculation in R:

```
> sepe <- read.table("https://stat.ethz.ch/Teaching/Datasets/cas-das/sepe.dat",
                     header = TRUE)
> attach(sepe)

> alpha <- 0.05
> rho <- 0.1
> N <- NROW(sepe)
> CV_y <- sd(expend) / mean(expend)
> z <- qnorm(1 - alpha / 2)
```

Ignoring the finite population correction,

```
> n0 <- (z * CV_y / rho)^2
> ceiling(n0)
```

```
[1] 242
```

Including the finite population correction,

```
> n <- n0 / (1 + n0 / N)
> ceiling(n)
```

```
[1] 186
```

Hence, the sample should contain at least 186 units.

b) Use the minimum sample size from part (a) to estimate total environmental protection expenditures and report a 95% confidence interval. Cochran proposed the rule $n > 25G_1^2$, where G_1 is Fisher's coefficient of skewness, i.e., the third standardized moment.[1] Sugden et al. proposed the modified rule $n > 28 + 25G_1^2$.[2]

Does the sample size satisfy Cochran's rule? Does it satisfy the rule proposed by Sugden et al.? What do these rules suggest about the adequacy of the 95% confidence interval?

> **Hint.** You can use the following function to compute the finite-population moment coefficient of skewness.
>
> ```
> skewness_pop <- function(x)
> {
>     z <- x - mean(x)
>     mean(z^3) / mean(z^2)^(3 / 2)
> }
> ```

Solution. The skewness of `expend` is approximately

```
> G1 <- skewness_pop(expend)
> G1
```

```
[1] 2.276473
```

Cochran's rule requires

$$n > 25G_1^2,$$

which gives

```
> ceiling(25 * G1^2)
```

```
[1] 130
```

Sugden et al.'s rule for a Studentized confidence interval is

$$n > 28 + 25G_1^2,$$

giving

[1] Cochran, W. G. (1977). Sampling Techniques. 3rd ed., New York: John Wiley & Sons, p. 42.
[2] Sugden, R. A., Smith, T. M. F., and Jones, R. P. (2000). Cochran's Rule for Simple Random Sampling. Journal of the Royal Statistical Society Series B, 62, pp. 787–793.

```
> ceiling(28 + 25 * G1^2)
```

```
[1] 158
```

The planned sample size of 186 satisfies both rules. Thus, these rules support using the usual normal-based 95 % confidence interval. They do not, however, guarantee an exact 95 % coverage probability.

c) Under the same assumptions as in part a), determine the sample size so that the coefficient of variation of the estimator

$$\mathsf{CV}_p(\widehat{t}_y) = \frac{\mathsf{SE}_p(\widehat{t}_y)}{\mathbb{E}_p(\widehat{t}_y)}$$

is at most 5 %, where $\widehat{t}_y = N\bar{y}$.

Solution. Under SRS,

$$\mathsf{var}_p(\widehat{t}_y) = N^2\left(1 - \frac{n}{N}\right)\frac{\sigma_y^2}{n}, \qquad \mathbb{E}_p(\widehat{t}_y) = t_y = N\bar{y}_U.$$

Hence,

$$\mathsf{CV}_p(\widehat{t}_y) = \mathrm{CV}_y\sqrt{\frac{1 - n/N}{n}}.$$

We require

$$\mathsf{CV}_p(\widehat{t}_y) \leq 0.05.$$

Ignoring the finite-population correction first gives

$$\frac{\mathrm{CV}_y}{\sqrt{n}} \leq 0.05,$$

or equivalently,

$$n \geq n_0 = \left(\frac{\mathrm{CV}_y}{0.05}\right)^2.$$

Accounting for the finite-population correction gives

$$n \geq \frac{n_0}{1 + n_0/N}.$$

Thus, the minimum sample size is the smallest integer satisfying this inequality.

Calculation in R:

```
> n0 <- (CV_y / 0.05)^2
> ceiling(n0)
```

```
[1] 252
```

```
> n <- n0 / (1 + n0 / N)
> ceiling(n)
```

```
[1] 192
```

Thus, at least 192 units are required.

d) How large must the sample be to estimate the population proportion P_{31} of establishments in industry class 31 (indicator W_{31}) with a standard error of at most 5 percentage points? Use

$$\widehat{P}_{31} = \frac{1}{n} \sum_{i \in s} \mathbb{1}\{i \in W_{31}\},$$

where

$$\mathbb{1}\{i \in W_{31}\} = \begin{cases} 1 & \text{if } i \text{ belongs to industry class 31,} \\ 0 & \text{otherwise.} \end{cases}$$

Solution. From Problem 1.2c), $\widehat{P}_{31} = 0.1725$. Let $\mathbb{1}_{31}$ denote the indicator variable defined above. Under SRS,

$$\mathsf{SE}_p(\widehat{P}_{31}) = \sqrt{\left(1 - \frac{n}{N}\right) \frac{\sigma^2{}_{\mathbb{1}_{31}}}{n}},$$

where $\sigma^2{}_{\mathbb{1}_{31}}$ is the finite-population variance of the indicator.

Ignoring the finite population correction first,

$$n_0 = \frac{\sigma^2{}_{\mathbb{1}_{31}}}{0.05^2}.$$

The corrected sample size is

$$n = \frac{n_0}{1 + n_0/N}.$$

Calculation in R:

```
> I31 <- code
> P31 <- mean(I31)
> sigma2_I31 <- var(I31)
> n0 <- sigma2_I31 / 0.05^2
> ceiling(n0)

[1] 58

> n <- n0 / (1 + n0 / N)
> ceiling(n)

[1] 54
```

Thus, the sample should contain at least 54 establishments.

Problem 2.2 (Simple random sampling). We use the data set `sepe.dat` from Problem Set 1.

a) Draw a simple random sample without replacement (SRS) of size $n = 200$ from the `sepe.dat` population, which is of size $N = 800$. Estimate

   i) the mean sepeironmental protection expenditure per establishment, $\bar{y}_U$;

   ii) total expenditure, t_y;

iii) mean expenditure per employee among establishments with at least one employee,

$$\bar{q}_{U+} = \frac{1}{N_+} \sum_{i \in U_+} \frac{y_i}{x_i}, \qquad U_+ = \{i \in U : x_i > 0\}.$$

Here, y_i denotes expenditure and x_i the number of full-time employees of establishment i. Also estimate the standard error of each estimator.

Compare the estimated standard errors with the corresponding population-based standard errors. For the sample mean, also compare the estimated variance with the Monte Carlo (MC) variance from Problem 1.3a).

> **Hint.** Write down the three estimators before doing the calculations in R. To reproduce the solution, use `set.seed(11)` before drawing the sample. A sample of indices can then be obtained with
>
> ```
> N <- NROW(sepe)
> in_sample <- sample(N, 200)
> sepe_sample <- sepe[in_sample, ]
> ```

Solution. First load the population and draw the sample.

```
> sepe <- read.table("https://stat.ethz.ch/Teaching/Datasets/cas-das/sepe.dat",
                     header = TRUE)

> N <- NROW(sepe)
> set.seed(11)
> n <- 200
> in_sample <- sample(N, n)
> sepe_sample <- sepe[in_sample, ]
> fpc <- 1 - n / N
```

i) **Mean expenditure.** The estimator and its estimated standard error are

```
> m_y <- mean(sepe_sample$expend)
> m_y

[1] 82508.45

> se_m_y <- sqrt(fpc * var(sepe_sample$expend) / n)
> se_m_y

[1] 3964.578
```

Using the complete finite population, the standard error is

```
> SE_m_y <- sqrt(fpc * var(sepe$expend) / n)
> SE_m_y

[1] 4170.373
```

The Monte Carlo variance from Problem 1.3a) was 17479711; thus, the MC estimate of standard error is

```
> sqrt(17479711)
```

```
[1] 4180.874
```

The two standard errors are very close.

ii) **Total expenditure.** Since $\widehat{t}_y = N\bar{y}$, estimate, (theoretical) standard error, and estimated standard error are:

```
> t_y <- N * m_y
> t_y
```

```
[1] 66006760
```

```
> se_t_y <- N * se_m_y
> se_t_y
```

```
[1] 3171662
```

```
> SE_t_y <- N * SE_m_y
> SE_m_y
```

```
[1] 4170.373
```

iii) **Mean expenditure per employee.** Restrict attention to establishments with at least one employee. Conditional on the number n_+ of such establishments in the sample, they form an SRS from U_+. Thus, the usual SRS variance formula applies within this subpopulation.

Restrict data and variables to cases with at least one employee:

```
> sepe_pos <- sepe[sepe$empl > 0, ]
> sepe_sample_pos <- sepe_sample[sepe_sample$empl > 0, ]
> N_pos <- NROW(sepe_pos)
> n_pos <- NROW(sepe_sample_pos)
```

Define the expenditure-per-employee variable for establishments with at least one employee:

```
> q <- sepe_sample_pos$expend / sepe_sample_pos$empl
> q_pop <- sepe_pos$expend / sepe_pos$empl
```

```
> fpc_pos <- 1 - n_pos / N_pos
```

The estimator and its estimated standard error are

```
> m_q <- mean(q)
> m_q
```

```
[1] 1151.237
```

```
> se_m_q <- sqrt(fpc_pos * var(q) / n_pos)
> se_m_q
```

```
[1] 148.2974
```

The corresponding population-based standard error is

```
> SE_m_q <- sqrt(fpc_pos * var(q_pop) / n_pos)
> SE_m_q
```

```
[1] 176.0603
```

b) Repeat the three estimates from part a) using the R package `survey` and the sample `sepe_sample`.

Hint. Attach the `survey` package to the search path, add the population size N to the sample data, and define the survey design object `dn_sepe`:

```
library("survey")
sepe_sample$N <- N
dn_sepe <- svydesign(ids = ~1, fpc = ~N, data = sepe_sample)
```

The argument `ids = ~1` specifies a design without clustering. Supplying `fpc = ~N` incorporates sampling without replacement from a population of size N.

New variables can be added to a design object with `update()`. A design object can be restricted to a subpopulation using `subset()`.

Use `svymean()` to estimate population means and `svytotal()` to estimate population totals. Their basic syntax is

```
svymean(~variable, design)
svytotal(~variable, design)
```

The first argument is an R formula specifying the variable to be analyzed. The tilde `~` is part of the formula syntax; for example, `~expend` specifies the variable `expend`. The second argument is the survey design object containing the sample data and the information about the sampling design.

The functions return objects containing both the estimates and information about their sampling variability. The corresponding methods `coef()`, `SE()`, and `cv()` extract the point estimates, standard errors, and coefficients of variation, respectively.

Solution. Define the survey design.

```
> library("survey")
> sepe_sample$N <- N
> dn_sepe <- svydesign(ids = ~1, fpc = ~N, data = sepe_sample)
```

The mean and total expenditure are obtained directly:

```
> svymean(~expend, dn_sepe)

        mean     SE
expend 82508 3964.6

> svytotal(~expend, dn_sepe)

          total      SE
expend 66006760 3171662
```

For mean expenditure per employee, restrict the design to establishments with at least one employee and then define the ratio.

```
> dn_pos <- subset(dn_sepe, empl > 0)
> dn_pos <- update(dn_pos, q = expend / empl)
> svymean(~q, dn_pos)
```

```
     mean     SE
q 1151.2 148.14
```

The estimates agree with part a), apart from possible differences in the variance calculation for the domain mean.

c) Using the design from part b), construct a 95 % Wald and a logit-based confidence interval for the population proportion P_{31} of establishments in industry class 31, identified by `code = 1`. Use `svyciprop()`.

> **Hint.** A normal-based Wald interval on the probability scale is obtained with
>
> *`svyciprop(~code, dn_sepe, method = "mean", level = 0.95, df = Inf)`*
>
> Setting `df = Inf` specifies that the confidence interval should use the standard normal distribution rather than a Student t-distribution based on the design degrees of freedom.
>
> Using `method = "logit"` instead gives a logit-based confidence interval. The logit interval respects the natural bounds of a proportion and can better accommodate asymmetry when the estimated proportion is close to 0 or 1.

Solution. The Wald interval is

```
> ci_p31 <- svyciprop(~code, dn_sepe, method = "mean", level = 0.95, df = Inf)
> ci_p31

            2.5% 97.5%
code 0.170 0.125 0.215
```

For comparison, a logit-based interval is

```
> svyciprop(~code, dn_sepe, method = "logit", level = 0.95, df = Inf)

            2.5% 97.5%
code 0.170 0.129 0.220
```

The two intervals need not be identical because they use different transformations.

Problem 2.3 (Stratified simple random sampling). Use the file `sepe.dat` again. Define the stratum indicator `strat = 1` for establishments with `empl` ≤ 250 and `strat = 2` otherwise. Also define `Nh`, which gives the population size of the corresponding stratum for each establishment. Add both variables to the data set.
Draw a stratified simple random sample without replacement, with $n_1 = 100$ establishments from the first stratum and $n_2 = 50$ from the second stratum. Estimate the population mean of environmental expenditures and its standard error.
Finally, compare the estimated standard error with that from a simple random sample of $n = 150$ establishments. For the simple random sample, compute the standard error both with and without the finite population correction.

> **Hint.** Load the data `sepe.dat` data and define `strat` and `Nh` as described above. Attach the `survey` package to the search path and draw the stratified sample:

```
library("survey")
set.seed(11)
sepe_sample <- sepe[stratsample(sepe$strat, c("1" = 100, "2" = 50)), ]
```

The function `stratsample()` is provided by the `survey` package. To reproduce the solution, use `set.seed(11)` before drawing the sample. Next, define the survey design object by

```
dn_sepe <- svydesign(ids = ~1, strata = ~strat, fpc = ~Nh,
                     data = sepe_sample)
```

Then use `svymean()` to estimate the population mean and `SE()` to extract its estimated standard error.

Solution.

```
> sepe <- read.table("https://stat.ethz.ch/Teaching/Datasets/cas-das/sepe.dat",
                     header = TRUE)
```

Define the strata and their population sizes:

```
> sepe$strat <- ifelse(sepe$empl <= 250, 1, 2)
> Nh <- table(sepe$strat)
> sepe$Nh <- as.numeric(Nh[as.character(sepe$strat)])
> Nh

  1   2
580 220
```

Draw the stratified sample:

```
> library("survey")
> set.seed(11)
> idx <- stratsample(sepe$strat, c("1" = 100, "2" = 50))
> sepe_sample <- sepe[idx, ]
> table(sepe_sample$strat)

  1   2
100  50
```

Define the design and estimate the population mean:

```
> dn_sepe <- svydesign(ids = ~1, strata = ~strat, fpc = ~Nh,
                       data = sepe_sample)
> m_y <- svymean(~expend, dn_sepe)
> m_y

        mean     SE
expend 86472 4019.5
```

```
> SE(m_y)

         expend
expend 4019.504
```

Thus, the estimated standard error of the stratified estimator is 4'020.

Remarks. For comparison, we draw a separate SRS of size 150:

```
> set.seed(11)
> srs_sample <- sepe[sample(seq_len(nrow(sepe)), size = 150), ]
```

We add population size to `srs_sample` and define the design object:

```
> srs_sample$N <- 800
> dn_srs <- svydesign(ids = ~1, fpc = ~N, data = srs_sample)
```

Ror the SRS sample, the estimated std. err. of the mean is

```
> SE(svymean(~expend, dn_srs))

         expend
expend 4729.543
```

which is considerably large than the estimated std. err. in case of the stratified sample, which is (approximately) 4'020.